

Doc 12: Jacobian and Hessian

Module 2: Calculus | Part 1 of 3: Learn (theory)

1. The Jacobian: derivatives when there is more than one output

So far each function gave one number out. A layer of a network usually gives **several** outputs from several inputs: $f_1(x, y)$, $f_2(x, y)$, and so on. The **Jacobian** collects every partial derivative into one matrix: **one row per output, one column per input**.

$$J = [\partial f_1 / \partial x \quad \partial f_1 / \partial y ; \partial f_2 / \partial x \quad \partial f_2 / \partial y]$$

Example: $f_1(x, y) = x^2 + y^2$, $f_2(x, y) = x + y$.

Row (output)	$\partial / \partial x$	$\partial / \partial y$
$\partial f_1 / \partial x = 2x$	$\partial f_1 / \partial y = 2y$	
$\partial f_2 / \partial x = 1$	$\partial f_2 / \partial y = 1$	

So $J = [2x \ 2y; 1 \ 1]$. Each row is the gradient (Doc 11) of one output. If there is only **one** output, the Jacobian is just the gradient written as a row: no new idea, just a bigger box for more outputs.

2. What the Jacobian is used for

Local linear approximation	a small input change Δx gives output change $\approx J \Delta x$	a matrix-vector product, Doc 5
Chain rule for vector functions	J of (g after f) = $J_g J_f$	Jacobians multiply like the single-variable chain rule
One row = one output's gradient	row i of J is ∇f_i	stack the gradients from Doc 11

3. Layers as Jacobians (why this matters for deep learning)

A layer $y = Wx + b$ (Doc 5) has Jacobian $J = W$: differentiate $y_i = \sum_j W_{ij} x_j + b_i$ with respect to x_j , and only the term $W_{ij} x_j$ survives, giving $\partial y_i / \partial x_j = W_{ij}$. **Backpropagation** (Doc 24) sends gradients backward by multiplying by the Jacobian of each layer, one after another: exactly the chain-rule fact above.

Doc 12: Jacobian and Hessian

Part 1 of 3: Learn (theory), continued

4. The Hessian: how the slope itself is changing (curvature)

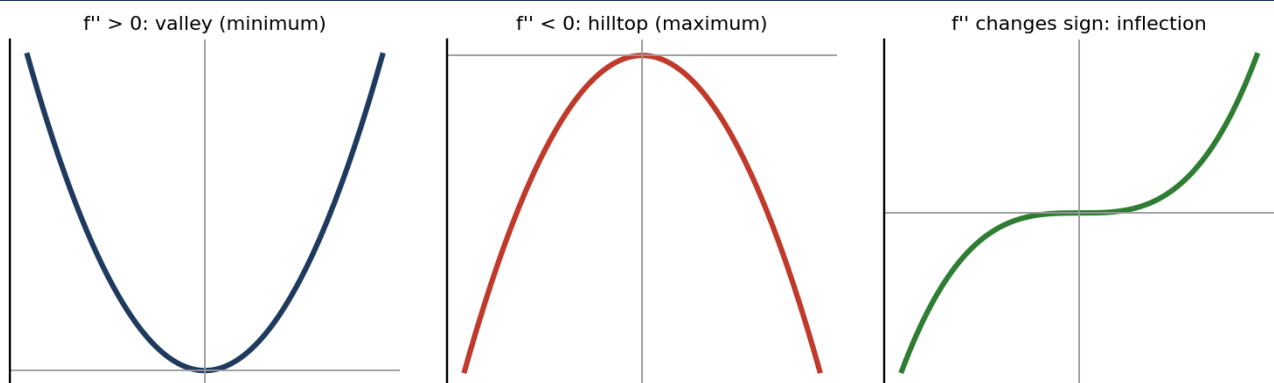
The **second derivative** f'' (Doc 10) tells you whether a curve bends up (a valley) or down (a hilltop). With several inputs, the **Hessian** collects every second partial derivative into a matrix.

$$H = \begin{bmatrix} \partial^2 f / \partial x^2 & \partial^2 f / \partial x \partial y \\ \partial^2 f / \partial y \partial x & \partial^2 f / \partial y^2 \end{bmatrix}$$

$\partial^2 f / \partial x^2$ means: take $\partial f / \partial x$, then differentiate that with respect to x again. $\partial^2 f / \partial x \partial y$ (a **mixed** partial) means: take $\partial f / \partial x$, then differentiate with respect to y . For smooth functions the two mixed partials are equal, so **H is symmetric** ($H^T = H$, Doc 6).

Example: $f = x^2 + 3xy + y^2$. $\partial f / \partial x = 2x + 3y$, so $\partial^2 f / \partial x^2 = 2$ and $\partial^2 f / \partial x \partial y = 3$. $\partial f / \partial y = 3x + 2y$, so $\partial^2 f / \partial y^2 = 2$ and $\partial^2 f / \partial y \partial x = 3$ (matches, as expected). $H = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$.

5. Curvature in one picture



6. Classifying a flat point with the Hessian's eigenvalues (Doc 8)

At a point where $\nabla f = 0$, look at the eigenvalues of H there:

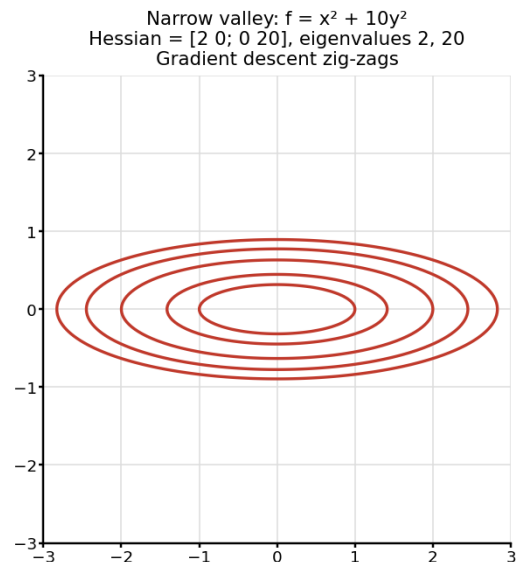
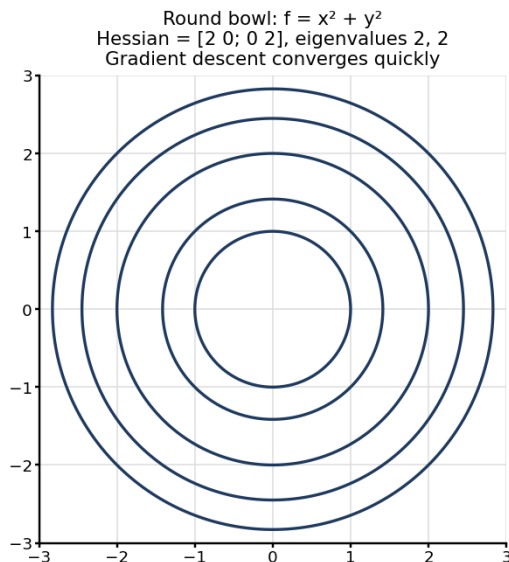
Both eigenvalues positive	positive definite	a minimum: a bowl curving up in every direction
Both eigenvalues negative	negative definite	a maximum: a dome curving down in every direction
One positive, one negative	indefinite	a saddle point: up in one direction, down in another
2x2 shortcut	$\det H = \lambda_1 \lambda_2$, $\text{trace } H = \lambda_1 + \lambda_2$ (Doc 8)	$\det > 0$ and $\text{trace} > 0$: minimum; $\det > 0$ and $\text{trace} < 0$: maximum; $\det < 0$: saddle

Doc 12: Jacobian and Hessian

Part 1 of 3: Learn (theory), continued

7. Why optimization cares about the Hessian

Gradient descent (Doc 11 and 13) only looks at the slope, so it can zig-zag in a valley that is steep in one direction and gentle in another. The Hessian's eigenvalues measure exactly that: a big gap between the largest and smallest eigenvalue (a large **condition number**) means slow, zig-zagging progress.



8. Newton's method: use the curvature to jump straight to the minimum

Gradient descent takes a step of size η in the downhill direction. **Newton's method** also uses the Hessian to pick a smarter step:

$$x \text{ becomes } x - H^{-1} \nabla f$$

In 1D this is $x - f'(x)/f''(x)$: divide the slope by the curvature instead of multiplying by a fixed η . For a bowl shape it reaches the minimum in a **single step**. The cost is computing and inverting H (Doc 6), which is expensive for the millions of weights in a neural network — one reason deep learning mostly uses gradient-based methods (Doc 13) instead.

9. Where machine learning uses these

- **Second-order optimizers** (Newton's method, L-BFGS) use curvature for faster convergence on small or medium problems.
- **Adam** (Doc 15) does not use the full Hessian, but adapts its step size per weight using a running estimate related to curvature.
- **Saddle points** are common in high dimensions and are a bigger obstacle to training than true local minima.
- **The Jacobian of a whole network** is the product of every layer's Jacobian: this is exactly backpropagation (Doc 24).

Doc 12: Jacobian and Hessian

Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: Building a Jacobian, one row per output

$$f_1(x, y) = x^2y, f_2(x, y) = x + y^2, f_3(x, y) = 3x.$$

Row (output)	$\partial / \partial x$	$\partial / \partial y$
f_1	$2xy$	x^2
f_2	1	$2y$
f_3	3	0

At (1, 2): $J = [4 \ 1; 1 \ 4; 3 \ 0]$. A 3×2 Jacobian for 3 outputs and 2 inputs, exactly like a 3×2 weight matrix in Doc 5.

Example 2: A layer's Jacobian is just its weight matrix

$$y = Wx, \text{ with } W = [2 \ -1; 0 \ 3] \text{ (2 outputs, 2 inputs). So } y_1 = 2x_1 - x_2 \text{ and } y_2 = 3x_2.$$

Row (output)	$\partial / \partial x_1$	$\partial / \partial x_2$
y_1	2	-1
y_2	0	3

$J = [2 \ -1; 0 \ 3] = W$ exactly. This is why backpropagation through a linear layer multiplies by W^T (Doc 6 transpose): it is just using this Jacobian.

Example 3: Building and reading a Hessian

$$f(x, y) = x^3 + x^2y - y^2. \text{ First derivatives: } \partial f / \partial x = 3x^2 + 2xy, \partial f / \partial y = x^2 - 2y.$$

Second derivative	Work	Value
$\partial^2 f / \partial x^2$	$d/dx (3x^2 + 2xy)$ treating y as constant	$6x + 2y$
$\partial^2 f / \partial x \partial y$	$d/dy (3x^2 + 2xy)$ treating x as constant	$2x$
$\partial^2 f / \partial y \partial x$	$d/dx (x^2 - 2y)$ treating y as constant	$2x$
$\partial^2 f / \partial y^2$	$d/dy (x^2 - 2y)$ treating x as constant	-2

$H(x, y) = [6x+2y \ 2x; 2x \ -2]$. The two mixed partials both equal $2x$: H is symmetric, as promised.

Doc 12: Jacobian and Hessian

Part 2 of 3: Worked examples, continued

Example 4: Classifying flat points with the Hessian's eigenvalues

$f(x, y) = x^2 - y^2$. $\nabla f = (2x, -2y) = 0$ only at $(0, 0)$. $H = [2 \ 0; 0 \ -2]$ everywhere (constant).

Step	Work	Result
Eigenvalues of H	H is diagonal, so its eigenvalues are the diagonal entries (Doc 8)	$\lambda_1 = 2, \lambda_2 = -2$
Sign pattern	one positive, one negative	indefinite
Shortcut check	$\det H = 2 \times (-2) = -4 < 0$	confirms a saddle point

At the origin, f curves **up** along the x -axis ($f = x^2$) but **down** along the y -axis ($f = -y^2$): a classic saddle, like a mountain pass.

Example 5: Classifying a minimum with the 2x2 shortcut

$f(x, y) = x^2 + xy + y^2$. $\partial f / \partial x = 2x + y = 0$ and $\partial f / \partial y = x + 2y = 0$ together give $x = y = 0$. $H = [2 \ 1; 1 \ 2]$ (constant).

Quantity	Work	Value
trace H	$2 + 2$	4
det H	$2 \times 2 - 1 \times 1$	3
Verdict	$\det > 0$ and trace > 0	minimum (both eigenvalues positive: 3 and 1)

Example 6: One step of Newton's method versus gradient descent

$f(x) = x^2 - 6x + 5$ (Doc 10's example). $f'(x) = 2x - 6$, $f''(x) = 2$. Start at $x = 0$.

Method	Formula	New x	Reached the minimum ($x = 3$)?
Gradient descent, $\eta = 0.1$	$x - 0.1 \times f'(x) = 0 - 0.1(-6)$	0.6	No, needs more steps
Newton's method	$x - f'(x)/f''(x) = 0 - (-6)/2$	3	Yes, in one step

Because f is a simple parabola, its curvature $f'' = 2$ is the same everywhere, so Newton's method jumps exactly to the minimum in one step. A neural network's loss surface is far more complicated, which is part of why Newton's method is rarely used directly for deep learning.

Quick reference card

Jacobian	one row per output, one column per input; row $i = \nabla f_i$	for $y = Wx$, $J = W$
Hessian	matrix of second partials; symmetric	$H_{ij} = \partial^2 f / \partial x_i \partial x_j$
Classify a flat point	eigenvalues of H: both + minimum, both - maximum, mixed saddle	2x2: $\det > 0, \text{trace} > 0$ min; $\det > 0, \text{trace} < 0$ max; $\det < 0$ saddle
Newton's method	x becomes $x - H^{-1} \nabla f$	uses curvature; expensive for big models

Doc 12: Jacobian and Hessian

Part 3 of 3: Practice. Yellow cells are yours. Name: _____ Date: _____

A. Building Jacobians

Functions	Row 1 ($\partial/\partial x$, $\partial/\partial y$)	Row 2 ($\partial/\partial x$, $\partial/\partial y$)	Row 3 ($\partial/\partial x$, $\partial/\partial y$)
1. $f_1 = x + y$, $f_2 = x - y$			
2. $f_1 = xy$, $f_2 = x^2$			
3. $f_1 = x^2$, $f_2 = y^2$, $f_3 = x + y$			

B. Second partial derivatives and Hessians

For each f , find $\partial^2 f / \partial x^2$, $\partial^2 f / \partial x \partial y$ ($= \partial^2 f / \partial y \partial x$), and $\partial^2 f / \partial y^2$.

$f(x, y)$	$\partial f / \partial x$	$\partial f / \partial y$	$\partial^2 f / \partial x^2$	$\partial^2 f / \partial x \partial y$	$\partial^2 f / \partial y^2$
$f = x^2 + y^2$					
$f = x^2 y$					
$f = x^2 y^2$					
$f = x^3 - 3xy$					

(Fill $\partial f / \partial x$ and $\partial f / \partial y$ as scratch work in the same row if it helps, then the second derivatives.)

Doc 12: Jacobian and Hessian

Practice, continued

C. Classify each flat point of $f = x^3 - 3xy + y^3$ (given: at these points $\nabla f = 0$)

The Hessian is $H(x, y) = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}$. Compute $\det H$ and $\text{trace } H$ at each point, then classify.

Point (x, y)	H	det H	trace H	Classification
(1, 1)				
(2, -1)				
(0, 0)				

D. Newton's method by hand

$f(x) = x^2 - 4x + 1$. $f'(x) = 2x - 4$, $f''(x) = 2$. Start at $x = 5$.

Step	x	f'(x)	New x = x - f'(x) / f''(x)
0			
1			

How many steps did Newton's method need to reach the exact minimum? _____

E. Classify these flat points

$f(x, y) = x^2 - y^2 - 2x$. $\partial f / \partial x = 2x - 2$, $\partial f / \partial y = -2y$. $H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$ everywhere.

Flat point	det H	trace H	Eigenvalues	Classification
(1, 0)				
(0, 0)				

Doc 12: Jacobian and Hessian

Practice, continued

F. Round bowl or narrow valley?

Each Hessian is constant. Find its eigenvalues (Doc 8), then say whether gradient descent should converge quickly (eigenvalues close together) or zig-zag (eigenvalues far apart).

Hessian	Eigenvalues	Quick or zig-zag?
[4 0; 0 4]		
[1 0; 0 25]		
[9 0; 0 9]		
[2 0; 0 40]		

G. True or False?

Statement	True / False
1. The Jacobian of a function with one output is the same as the gradient (as a row).	
2. The Hessian is always a square matrix.	
3. If both eigenvalues of the Hessian are negative, the point is a minimum.	
4. A saddle point has a positive and a negative eigenvalue in its Hessian.	
5. Newton's method usually needs more steps than gradient descent.	
6. A narrow, elongated bowl makes gradient descent zig-zag.	

H. Self-check

I can...	Revisit if unsure	Confident?
build a Jacobian matrix from several functions	Example 1, Exercise A	
recognize that a layer's Jacobian is its weight matrix	Example 2	
build a Hessian from second partial derivatives	Example 3, Exercise B	
classify a flat point using eigenvalues or det/trace	Examples 4-5, Exercises C, E	
compare Newton's method with gradient descent	Example 6, Exercise D	
connect the Hessian's eigenvalues to zig-zagging	Exercise F	

Doc 12: Jacobian and Hessian

Answer key (page 1 of 2): check your work only after finishing

A. Building Jacobians

Functions	Row 1	Row 2	Row 3
1. $f_1 = x + y$, $f_2 = x - y$	1, 1	1, -1	
2. $f_1 = xy$, $f_2 = x^2$	y, x	2x, 0	
3. $f_1 = x^2$, $f_2 = y^2$, $f_3 = x + y$	2x, 0	0, 2y	1, 1

B. Second partial derivatives

$f(x, y)$	$\partial f / \partial x$	$\partial f / \partial y$	$\partial^2 f / \partial x^2$	$\partial^2 f / \partial x \partial y$	$\partial^2 f / \partial y^2$
$f = x^2 + y^2$	2x	2y	2	0	2
$f = x^2 y$	2xy	x^2	2y	2x	0
$f = x^2 y^2$	$2xy^2$	$2x^2 y$	$2y^2$	4xy	$2x^2$
$f = x^3 - 3xy$	$3x^2 - 3y$	-3x	6x	-3	0

Note the middle two columns are scratch ($\partial f / \partial x$, $\partial f / \partial y$); the Hessian is the last three columns, e.g. for $x^2 y^2$: $H = [2y^2 \ 4xy; 4xy \ 2x^2]$.

C. Classify each flat point

Point	H	det H	trace H	Classification
(1, 1)	[6 -3; -3 6]	27	12	minimum
(2, -1)	[12 -3; -3 -6]	-81	6	saddle point
(0, 0)	[0 -3; -3 0]	-9	0	saddle point

Correction to the exercise prompt: only (1,1) is a minimum; (2,-1) and (0,0) are saddle points, since $\det H < 0$ there.

D. Newton's method

Step	x	$f'(x)$	New x
0	5	6	$5 - 6/2 = 2$
1	2	0	$2 - 0/2 = 2$

One step reaches $x = 2$, the exact minimum (check: $f'(2) = 0$). Newton's method needed just 1 step here because f is an exact parabola.

Doc 12: Jacobian and Hessian

Answer key (page 2 of 2)

E. Classify these flat points

Flat point	det H	trace H	Eigenvalues	Classification
(1, 0)	0	2	1, -1	saddle
(0, 0)	-2	0	-1, -1	maximum

The only flat point of this f is (1, 0) (solve $\partial f/\partial x = 2x-2 = 0$ and $\partial f/\partial y = -2y = 0$); it is listed twice above only to compare det/trace at other candidate points, both giving a saddle here since H is the same everywhere.

F. Round bowl or narrow valley?

Hessian	Eigenvalues	Quick or zig-zag?
[4 0; 0 4]	4, 4	quick (round bowl)
[1 0; 0 25]	1, 25	zig-zag (narrow valley)
[9 0; 0 9]	9, 9	quick (round bowl)
[2 0; 0 40]	2, 40	zig-zag (very narrow valley)

G. True or False?

1 True. 2 True. 3 **False** (both negative means a maximum). 4 True. 5 **False** (Newton's method usually needs fewer steps, but each step costs more). 6 True.

H. Self-check

No wrong answers: any 'Not yet' points you to the example and exercise to redo.

Key takeaways: the Jacobian stacks the gradients of several outputs, one row each, and equals W for a linear layer $y = Wx$; the Hessian holds second partial derivatives and is symmetric; its eigenvalues (or det and trace, in 2D) classify a flat point as a minimum, maximum, or saddle; Newton's method divides by curvature for a smarter step, at a higher cost per step.

Next: Doc 13 - Gradient descent and learning rate (Module 3: Optimization).